

# gmpfrxx\_mkII

A header-only C++17 expression-template wrapper for GMP, MPFR, and GNU MPC

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## Abstract

**gmpfrxx\_mkII** provides value-semantic C++ classes for arbitrary-precision integers, rationals, real floating-point numbers, and complex numbers, layered on the GNU multiple-precision libraries GMP, MPFR, and GNU MPC. An expression-template arithmetic layer removes most temporaries from compound expressions. The wrapper is designed as numerical infrastructure: each floating wrapper has exactly the size and alignment of its backend C object, the default-state policy is explicit and predictable across threads and shared libraries, and a small set of opt-in compile-time fast paths is available for BLAS-like kernels. This manual documents the public API, the precision and default-state model, the expression-template lifetime contract, and the behavioral guarantees that callers can rely on.

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# 1 Introduction

`gmpfrxx_mkII` is a header-only C++17 wrapper library for GMP, MPFR, and GNU MPC. In its default mode it requires no separate compiled library: including a header and linking the relevant backend C libraries is sufficient. It exposes four numeric domains.

- GMP-backed exact integers (`mpz_class`) and rationals (`mpq_class`).
- GMP-backed binary floating point (`mpf_class`) and an MPF-backed complex type (`mpfc_class`) built from two `mpf_class` components.
- MPFR-backed correctly-rounded real floating point (`mpfr_class`).
- GNU MPC-backed complex floating point (`mpc_class`).

## 1.1 A from-scratch implementation

This is a from-scratch implementation built directly on the GMP, MPFR, and MPC C APIs. It does not include `gmpxx.h`, does not link `libgmpxx`, and does not bridge to or wrap any existing C++ wrapper. The expression-template layer, the class designs, and the default-state machinery are all original to this project. The result is compatibility-oriented in spirit — familiar type names and operator behavior — but it is not a full `gmpxx.h` clone, and it deliberately diverges where a clearer or more predictable contract is worthwhile.

## 1.2 Design priorities

The library favors predictability over implicit convenience. The default precision is a fixed, documented value; assignment to an existing object never silently changes its precision; a single scalar constructor argument is always a *value*, never a precision selector; and the floating wrappers add no per-object metadata around the backend C object, so arrays of wrappers keep native stride. These choices target use as numerical infrastructure — for example dense linear-algebra kernels — rather than ad hoc scripting convenience.

## 1.3 Public and internal namespaces

The public API lives in two namespaces:

```
namespace gmpxx;    // GMP-only API
namespace mpfrxx;   // MPFR API, plus the MPC API when the MPC header is included
```

The implementation lives in an internal namespace that is not part of the public API:

```
namespace gmpfrxx_mkII::detail;
```

In particular, `namespace gmpfrxx` is *not* a public API namespace. The `gmpfrxx_mkII` name appears only as the internal `detail` namespace and as header and macro prefixes.

This manual assumes familiarity with the underlying C libraries. It describes what the wrapper guarantees, not the internals of GMP, MPFR, or MPC.

## 2 Installation and requirements

### 2.1 Toolchain

| Component | Requirement                                                                                         |
|-----------|-----------------------------------------------------------------------------------------------------|
| Compiler  | A conforming C++17 compiler (GCC, Clang, or MSVC).                                                  |
| GMP       | Required by every public header.                                                                    |
| MPFR      | Required by <code>mpfrxx_mkII.h</code> and the aggregator. Must be built with thread-local storage. |
| GNU MPC   | Required only for <code>mpfrxx::mpc_class</code> .                                                  |

### 2.2 The MPFR TLS requirement

The MPFR default precision, rounding mode, and exponent range are treated as the single source of truth and are read from `libmpfr`'s thread-local state. The project therefore requires an MPFR build with thread-local storage: a CMake configuration must verify that `mpfr_buildopt_tls_p()` exists and reports TLS enabled, and must fail otherwise. There is no fallback default-state implementation for a non-TLS MPFR build.

### 2.3 Compiling a program

```
#include <gmpfrxx_mkII.h>
#include <iostream>

int main() {
    mpfrxx::mpfr_class x = 2;
    mpfrxx::mpfr_class y = sqrt(x);    // 512-bit by default
    std::cout << y << '\n';
}
```

```
c++ -std=c++17 example.cpp -lmpc -lmpfr -lgmp
```

If only real MPFR arithmetic is needed, include `mpfrxx_mkII.h` and link `-lmpfr -lgmp`; GNU MPC is not required.

### 3 Headers and namespaces

Headers are split by required backend and link dependency so that a translation unit pays only for the backends it uses.

| Header                      | Public API                                                                                                | Link dependency                 |
|-----------------------------|-----------------------------------------------------------------------------------------------------------|---------------------------------|
| <code>gmpxx_mkII.h</code>   | <code>gmpxx::mpz_class</code> , <code>mpq_class</code> , <code>mpf_class</code> , <code>mpfc_class</code> | <code>-lgmp</code>              |
| <code>mpfrxx_mkII.h</code>  | <code>mpfrxx::mpfr_class</code> (+ <code>mpz_class</code> , <code>mpq_class</code> aliases)               | <code>-lmpfr -lgmp</code>       |
| <code>mpcxx_mkII.h</code>   | <code>mpfrxx::mpc_class</code>                                                                            | <code>-lmpc -lmpfr -lgmp</code> |
| <code>gmpfrxx_mkII.h</code> | aggregator of all of the above                                                                            | <code>-lmpc -lmpfr -lgmp</code> |

`gmpxx_mkII.h` is deliberately GMP-only. It does not include `<mpfr.h>` or `<mpc.h>`, references no `mpfr_*` or `mpc_*` symbol, and requires neither library at link time. The GMP complex type `gmpxx::mpfc_class` is implemented from two `mpf_class` components and stays in the GMP-only header rather than being split into a separate complex header.

`mpfrxx_mkII.h` requires GMP and MPFR but not GNU MPC. `mpcxx_mkII.h` includes `mpfrxx_mkII.h` automatically and layers the complex type on top.

Two namespaces are used. `gmpxx` holds the GMP-only types. `mpfrxx` holds the MPFR and MPC types and re-exports the GMP integer and rational types as aliases:

```
namespace mpfrxx {
    using ::gmpxx::mpz_class;
    using ::gmpxx::mpq_class;
}
```

### 4 Public types overview

| Type                            | Backend                    | Notes                                      |
|---------------------------------|----------------------------|--------------------------------------------|
| <code>gmpxx::mpz_class</code>   | GMP <code>mpz_t</code>     | Exact integer.                             |
| <code>gmpxx::mpq_class</code>   | GMP <code>mpq_t</code>     | Exact rational.                            |
| <code>gmpxx::mpf_class</code>   | GMP <code>mpf_t</code>     | Binary float; no IEEE NaN/Inf/signed-zero. |
| <code>gmpxx::mpfc_class</code>  | two <code>mpf_class</code> | GMP complex; no GNU MPC dependency.        |
| <code>mpfrxx::mpfr_class</code> | MPFR <code>mpfr_t</code>   | Correctly-rounded real float.              |
| <code>mpfrxx::mpc_class</code>  | GNU MPC <code>mpc_t</code> | Complex float.                             |

The two complex families are distinct by design. `gmpxx::mpfc_class` is MPF-backed; `mpfrxx::mpc_class` is MPC-backed. Mixed arithmetic between the families is intentionally rejected at compile time.

Because `mpf_class` and `mpfc_class` are GMP `mpf_t`-backed, they do not model IEEE signed zero, NaN, or infinity, and cannot reproduce signed-zero branch cuts. In particular `gmpxx::atan2(0,0)` returns `+0` and the four IEEE signed-zero cases are indistinguishable. Use `mpfr_class` or `mpc_class` when signed-zero branch behavior is part of the numeric contract.

## 5 Quick start

```
#include <gmpfrxx_mkII.h>
using namespace gmpxx;
using namespace mpfrxx;

mpz_class n = 1;
for (int k = 1; k <= 50; ++k) n *= k;           // 50!

mpq_class q("22/7");
q.canonicalize();                             // arithmetic-ready

mpf_class a(1.0, 256), b = sqrt(mpf_class(2, 256));
mpf_class c = a + b * b;                      // evaluated immediately

mpfr_class x = const_pi();                    // 512-bit pi
mpfr_class y = sin(x / 6);

mpc_class z(mpfr_class(0), mpfr_class(1));    // i
mpc_class w = exp(z * const_pi());           // ~ -1
```

## 6 Precision and default state

### 6.1 Initial defaults

The wrapper default floating precision is **512 bits** for MPF, MPFR, and MPC, unless overridden by environment variables (Section 6.5).

Assignment to an *existing* floating object preserves the left-hand-side precision in every domain. Changing a wrapper default affects only *subsequently constructed* objects.

### 6.2 GMP MPF default precision

GMP's `mpf_set_default_prec()` / `mpf_get_default_prec()` are process-global, not thread-local. `mpf_class` therefore does not use GMP's global default as its policy; it owns its own default. Default construction uses `mpf_init2(..., gmpxx::default_mpf_precision_bits())`.

Two explicit escape hatches expose the GMP global C state for code that intentionally wants it; these are *not* the wrapper default policy:

```
gmpxx::unsafe_get_gmp_global_mpf_precision_bits();
gmpxx::unsafe_set_gmp_global_mpf_precision_bits(bits);
```

#### 6.2.1 Default-context modes

`GMPXX_MKII_DEFAULT_CONTEXT_MODE` is a program-wide build setting. All translation units linked into one program must use the same value.

**Frozen-env mode (default).** Header-only; no provider object file; no header-owned mutable TLS. The default MPF precision is 512 bits unless overridden by `GMPXX_DEFAULT_MPF_PRECISION_BITS`, and the environment is read once and then frozen. Runtime mutation of the wrapper default is unsupported: `gmpxx::set_default_mpf_precision_bits(...)` is a compatibility no-op, and `gmpxx::reload_default_mpf_precision_bits_from_environment()` keeps the frozen value. Because the frozen value lives in a function-local static, each linked image may observe its own frozen default; this is acceptable for the simple header-only mode but is not a cross-DSO mechanism.

**External-provider mode.** Opt-in and not header-only: exactly one linked provider implementation owns the mutable per-thread default-context storage. Missing provider symbols are a link error. In this mode `gmpxx::set_default_mpf_precision_bits(...)` sets the calling thread's provider-owned default and `gmpxx::default_mpf_precision_guard` restores the previous value on scope exit. Do not embed the provider object in multiple shared libraries; designate a single owning library. Provider identity is checkable through `gmpxx_mkII_default_context_provider_token_v1()`, and all participating DSOs must observe the same token.

## 6.3 MPFR default state

MPFR default precision, rounding mode, and exponent range are owned by `libmpfr`'s thread-local state; the wrapper keeps no separate mutable storage for them. The public MPFR default API routes directly through MPFR:

```
mpfrxx::default_precision_bits();           // mpfr_get_default_prec()
mpfrxx::set_default_precision_bits(bits);   // mpfr_set_default_prec()
mpfrxx::default_rounding_mode();
mpfrxx::set_default_rounding_mode(rnd);
mpfrxx::default_emin(); mpfrxx::default_emax();
mpfrxx::set_default_exponent_range(emin, emax);
```

### 6.3.1 Per-thread first-use initialization

On a thread's first wrapper default access, if the MPFR default state still has its library-initial values (precision = 53, rounding = `MPFR_RNDN`, `emin/emax` at MPFR's defaults), the wrapper applies its 512-bit policy together with any `MPFRXX_*` environment overrides. If application code or another linked image has already changed any MPFR default field in that thread, the wrapper leaves the existing state alone. A DSO-local one-shot flag avoids repeatedly probing the defaults from hot paths; it stores no precision or rounding value of its own.

To establish an explicit boundary — for example before applying raw `mpfr_set_default_prec(...)` — call `mpfrxx::initialize_thread_defaults()` first. After the one-shot initialization has run, later raw MPFR default-state changes in the same thread are not overwritten by implicit wrapper access.

Default setters affect only the calling thread, and only subsequently constructed objects. `mpfrxx::reload_mpfr_defaults_from_environment()` reloads environment-derived defaults for the calling thread only. Applications that want consistent runtime defaults across worker threads must call a setter or reload API inside each thread.

## 6.4 MPC default state

MPC inherits the MPFR thread-local defaults: without an MPC-specific override, default real and imaginary precision equal the MPFR default precision, and default real and imaginary rounding equal the MPFR default rounding.

MPC-specific overrides affect *only* `mpc_class` defaults; they do not touch MPFR defaults. Precision and rounding may be symmetric or asymmetric:

```
mpfrxx::set_default_mpc_precision_bits(prec);           // symmetric
mpfrxx::set_default_mpc_precision_bits(real, imag);     // asymmetric
mpfrxx::set_default_mpc_rounding_mode(rnd);             // symmetric
mpfrxx::set_default_mpc_rounding_mode(real_rnd, imag_rnd); // asymmetric
```

Assigning a real value (an `mpfr_class`, `mpz_class`, `mpq_class`, or a supported scalar) to an `mpc_class` sets the imaginary component to +0, following GNU MPC's `mpc_set_fr`-family semantics.

## 6.5 Environment variables

```
GMPXX_DEFAULT_MPF_PRECISION_BITS

MPFRXX_DEFAULT_PRECISION_BITS
MPFRXX_DEFAULT_EMIN
MPFRXX_DEFAULT_EMAX
MPFRXX_DEFAULT_ROUNDING_MODE      # RNDN RNDZ RNDU RNDD RNDA RNDF

MPCXX_DEFAULT_PRECISION_BITS
MPCXX_DEFAULT_REAL_PRECISION_BITS
MPCXX_DEFAULT_IMAG_PRECISION_BITS
MPCXX_DEFAULT_ROUNDING_MODE
MPCXX_DEFAULT_REAL_ROUNDING_MODE
MPCXX_DEFAULT_IMAG_ROUNDING_MODE
```

Environment variables are process-startup configuration. Finalize all `GMPXX_*`, `MPFRXX_*`, and `MPCXX_*` variables before first wrapper default access and before worker threads begin numeric work. Concurrent `setenv/unsetenv/putenv` racing with wrapper default reads is outside the supported contract.

Invalid *user* environment values do not terminate the process: they are ignored with a diagnostic on `stderr` and the built-in default is used. By contrast, an external default-context provider with a mismatched ABI version or an internally inconsistent record may terminate the process after a diagnostic, because that indicates a binary-integration error rather than a configuration typo.

The `MPCXX_*` component variables override only their respective component; the other component inherits the corresponding MPC default or, failing that, the MPFR default. Asymmetric MPC environment precision and rounding are therefore valid.

## 7 Expression templates and the lifetime contract

Public arithmetic operators (+, -, \*, /, unary -) and compound assignments return expression nodes rather than evaluated values. A node may **borrow** an lvalue wrapper operand by raw reference and may **own** an rvalue operand.

This is a user-visible contract: **do not bind an expression to `auto` and let it outlive its operands.**

```
// BAD: e holds raw references to a and b.
auto e = a + b;
mutate_or_destroy(a);
gmpxx::mpf_class r = e;           // dangling if a was destroyed,
                                   // or stale if a was mutated.

// GOOD: evaluate immediately.
gmpxx::mpf_class r = a + b;

// GOOD: explicit snapshot when a stable value is needed.
auto snapshot = gmpxx::mpf_class(a + b);
```

Expression node types are `[[nodiscard]]`, so a discarded standalone expression such as `a + b`; produces a compiler diagnostic. This does not make a saved node lifetime-safe.

### 7.1 Generative random expressions

Random getters follow the same policy as upstream `gmpxx.h`: they return generator expression nodes, not value snapshots. This includes `get_z_bits`, `get_z_range`, `get_f` on `gmpxx::`

`gmp_randclass`, the matching getters on `mpfrxx::mpfr_randclass`, and the MPFR floating getters such as `get_fr(...)`. Evaluating such a node consumes the associated `gmp_randstate_t` and may produce a fresh value.

The number of times a random node embedded in an expression tree is generated is *not* a semantic contract. If one draw must be reused, materialize it first:

```
mpfrxx::mpfr_class r = state.get_fr(prec);
mpfrxx::mpfr_class y = r + r;           // reuses one draw
```

Random-state generation is not internally synchronized. Use a state from one thread at a time, or protect it with an external mutex.

## 8 Scalar policy and common types

Supported real scalar leaves are non-`bool` integral types up to 64 bits, `float`, and `double`. Integral leaves are normalized to signed or unsigned 64-bit storage before evaluation, so `long` and `unsigned long` are accepted through that normalization rather than through GMP's native `long` ABI. Supported scalars promote with `mpfr_class` to `mpfr_class` and with `mpc_class` to `mpc_class`.

`bool`, `wchar_t`, `char16_t`, `char32_t`, `long double`, and `__int128` are intentionally outside the scalar-leaf contract and must not become common-type promotion backdoors. (Direct `mpz_class` construction from `__int128` is supported where the compiler provides it, but `__int128` is still rejected as an expression scalar leaf.)

### 8.1 Explicit fixed-binary import adapters

Binary80 and binary128 name numeric formats, not stable C++ native types. The adapter layer therefore uses semantic wrappers parameterized by the consumer-selected native type:

```
auto r80 = gmpfrxx_mkII::adapters::make_binary80_source(value80);
auto r128 = gmpfrxx_mkII::adapters::make_binary128_source(value128);
mpfrxx::mpfr_class x80(r80);
mpfrxx::mpfr_class x128(r128);
```

The corresponding `make_binary80_complex_source(real, imag)` and `make_binary128_complex_source(real, imag)` wrappers import components without depending on extended `std::complex` support. Supported native types are selected by capability traits: `binary80` accepts a verified `long double` or `_Float64x` representation; `binary128` accepts `_Float128`, `__float128`, or a verified `binary128 long double`. Generic `long double` scalar promotion remains disabled, so the semantic wrapper is required.

When `mpcxx_mkII.h` or `gmpfrxx_mkII.h` is included, `std::complex<double>` is accepted as an MPC scalar operand. It promotes with `mpc_class` to `mpc_class`, and with `mpfr_class` to `mpc_class` because the result has a complex component. This support is MPC-only and does not make `mpfrxx_mkII.h` depend on GNU MPC.

## 9 Type reference

This section lists the principal public surface of each type. Constructors, accessors, and free functions follow GMP/MPFR/MPC naming conventions where a direct correspondence exists.

### 9.1 `gmpxx::mpz_class`

**Construction.** Default; copy/move; from `const char*` or `std::string` with an optional base (base 0 auto-detects); explicit from `double` (rejecting non-finite input) and from `mpz_srcptr`



; implicit from supported integral scalars (and from `__int128` where available); explicit from `mpf_class` and `mpq_class`; from integer/rational expression nodes. Construction from `bool` is deleted.

#### Accessors and queries.

```
std::string get_str(int base = 10) const;    // also to_string(base)
int set_str(const char* s, int base = 10);
double get_d() const;
signed long get_si() const; unsigned long get_ui() const;
std::int64_t get_i64() const; std::uint64_t get_u64() const;
explicit operator bool() const noexcept;
bool fits_sint_p() const; bool fits_slong_p() const; // etc.
const mpz_t& mpz_data() const noexcept;    // also get_mpz_t()
void swap(mpz_class&) noexcept;
```

**Number theory.** `abs`, `sqrt` (integer square root), `gcd`, `lcm`, `factorial`, `primorial`, `fibonacci`; `operator%` for integer remainder; pre/post increment and decrement; the `_mpz` literal suffix.

### 9.2 `gmpxx::mpq_class`

**Construction.** Default; copy/move; from string with optional base; from integer and rational expression nodes; pre/post increment and decrement; the `_mpq` literal suffix.

**Canonicalization.** See Section 12. The arithmetic boundary is enforced through:

```
void mpq_class::canonicalize();              // throws std::domain_error on 0 denom
bool mpq_class::has_zero_denominator() const;
gmpxx::mpq_canonicalize_checked(mpq_ptr);    // raw, throws on 0 denom
gmpxx::mpq_has_zero_denominator(mpq_srcptr);
```

### 9.3 `gmpxx::mpf_class`

#### Construction.

```
mpf_class();                                // default precision
mpf_class(double v);
mpf_class(double v, mp_bitcnt_t prec);
mpf_class(T integral_scalar [, mp_bitcnt_t prec]);
mpf_class(const mpz_class& [, mp_bitcnt_t prec]);
mpf_class(const mpq_class& [, mp_bitcnt_t prec]);
mpf_class(const char* s [, mp_bitcnt_t prec], int base = 10);
mpf_class(const std::string& s [, mp_bitcnt_t prec], int base = 10);
explicit mpf_class(mpf_srcptr v [, mp_bitcnt_t prec]);
mpf_class(const mpf_class& other, mp_bitcnt_t prec);
static mpf_class with_precision(mp_bitcnt_t prec);
static mpf_class with_precision(mp_bitcnt_t prec, double v);
```

Construction from `bool` is deleted.

#### Accessors.

```
mp_bitcnt_t precision() const noexcept;
void set_prec(mp_bitcnt_t prec);
double get_d() const;
signed long get_si() const; unsigned long get_ui() const;
const mpf_t& mpf_data() const noexcept;
void swap(mpf_class&) noexcept;
```

**Math (gmpxx namespace).** A self-contained MPF real-math set: `sqrt`, `sqr`, `abs`, `neg`, `ceil`, `floor`, `trunc`, `sgn`, `hypot`, `exp`, `log`, `log1p`, `expm1`, `sin`, `cos`, `tan`, `asin`, `acos`, `atan`, `atan2`, `sinh`, `cosh`, `tanh`, `asinh`, `acosh`, `atanh`, `pow`, `gamma`, `reciprocal_gamma`, remainder helpers, and the constants `const_pi`, `const_e`, `pi_over_two`, `pi_over_four`, `two_pi`, `log_two`, `log_ten`, `inv_log_two` (each available at the default precision or at an explicitly requested precision). Most functions also accept expression-node arguments. The `_mpf` literal suffix is provided.

#### 9.4 gmpxx::mpfc\_class

A GMP-only complex type built from two `mpf_class` components.

```
mpfc_class();
mpfc_class(const mpf_class& re, const mpf_class& im);
explicit mpfc_class(const mpf_class& re);           // im = 0
explicit mpfc_class(Scalar re);
mpfc_class(RealScalar re, ImagScalar im);
static mpfc_class with_precision(mp_bitcnt_t prec);
static mpfc_class with_precision(mp_bitcnt_t re_prec, mp_bitcnt_t im_prec);

const mpf_class& real() const noexcept;
const mpf_class& imag() const noexcept;
void real(const mpf_class&); void imag(const mpf_class&);
mp_bitcnt_t precision() const noexcept;
```

Complex math (`conj`, `real`, `imag`, `norm`, `abs`, `arg`, `polar`, `sqrt`, `exp`, `log`, the trigonometric and hyperbolic functions and their inverses, integer and floating `pow`) is provided in the `gmpxx` namespace. Because the backend is `mpf_t`, IEEE signed-zero branch cuts are not modeled.

#### 9.5 mpfrxx::mpfr\_class

##### Construction.

```
mpfr_class();                               // default precision
mpfr_class(double v [, mpfr_prec_t prec]);
mpfr_class(T integral_scalar [, mpfr_prec_t prec]);
mpfr_class(const gmpxx::mpz_class& [, mpfr_prec_t prec]);
mpfr_class(const gmpxx::mpq_class& [, mpfr_prec_t prec]);
explicit mpfr_class(const char* / const std::string&);
mpfr_class(const char* s, mpfr_prec_t prec, int base = 0);
explicit mpfr_class(mpfr_srcptr v [, mpfr_prec_t prec]);
mpfr_class(const mpfr_class& other, mpfr_prec_t prec);
```

Construction from `bool` is deleted.

##### Accessors and queries.

```
mpfr_prec_t precision() const noexcept;
void set_prec(mpfr_prec_t prec);
double get_d() const;
std::string get_str() const;
std::string get_str(mpfr_exp_t& exp, int base = 10, std::size_t ndig = 0) const;
std::string to_string(std::size_t ndig = 0) const;
const mpfr_t& mpfr_data() const noexcept; // also get_mpfr_t()
void swap(mpfr_class&) noexcept;
static mpfr_prec_t default_precision() noexcept;
```

**Eager math (mpfrxx namespace).** A broad correctly-rounded set, including the elementary functions (`sqrt`, `cbirt`, `exp`, `exp2`, `exp10`, `expm1`, `log`, `log2`, `log10`, `log1p`, the trigonometric,

hyperbolic, and unit/ $\pi$ -scaled trigonometric families and their inverses, `sec/csc/cot` and hyperbolic counterparts), special functions (`erf`, `erfc`, `gamma`, `lngamma`, `digamma`, `zeta`, `eint`, `li2`, the Bessel functions `j0/j1/ jn`, `y0/y1/yn`, `ai`), binary operations (`pow`, `atan2`, `hypot`, `fmod`, `agm`, `min`, `max`, `gamma_inc`, `beta`), fused operations (`fma`, `fmma`, `fmms`), scaling by powers of two (`mul_2ui`, `div_2ui`, `mul_2si`, `div_2si`), rounding to integers (`ceil`, `floor`, `trunc`), `setsign`, and the constants `const_pi`, `const_log2`, `const_e`, `const_log10`, `pi_over_two`, `const_euler`, `const_catalan`.

**Result precision of eager math.** For a multi-operand eager math function, the result precision is chosen from the MPFR object or MPFR-expression operands *before* scalar and exact GMP operands are materialized; those exact operands are then converted at the chosen precision and do not implicitly promote the result to the current default precision. If no MPFR operand contributes a positive precision, the wrapper default precision is used.

**Version-dependent functions.** Some MPFR functions are backend-version features (for example `powr`, the unit-based trigonometric functions, `rootn_si`, `compound_si`, `log2p1`, `exp2m1`, and `intmax` power helpers). The corresponding wrappers are exposed only when the included `<mpfr.h>` advertises the entry point; the rest of the MPFR wrapper remains available with older MPFR.

**Comparisons.** `mpfrxx::cmp(...)` is a total-order-style helper for ordered values; if either operand is NaN it throws `std::domain_error`. The equality, inequality, and ordering operators are NaN-aware predicates instead: equality is false for unordered operands, inequality is true, and ordering predicates are false when unordered.

## 9.6 `mpfrxx::mpc_class`

### Construction.

```
mpc_class();
mpc_class(const mpfr_class& re, const mpfr_class& im);
explicit mpc_class(const mpfr_class& re);           // im = +0
explicit mpc_class(const gmpxx::mpz_class& re);
explicit mpc_class(const gmpxx::mpq_class& re);
explicit mpc_class(Scalar re);
explicit mpc_class(const std::complex<double>& v);
mpc_class(RealScalar re, ImagScalar im);
explicit mpc_class(const char* / const std::string&);
mpc_class(const char* s, mpfr_prec_t prec, int base = 0);
mpc_class(const char* s, mpfr_prec_t re_prec, mpfr_prec_t im_prec, int base);
```

### Accessors.

```
mpfr_class real() const;      mpfr_class imag() const;
mpfr_prec_t real_precision() const noexcept;
mpfr_prec_t imag_precision() const noexcept;
mpfr_prec_t precision() const noexcept; // = real_precision()
std::string get_str(int base = 10, std::size_t ndig = 0) const;
std::string to_string(std::size_t ndig = 0) const;
const mpc_t& mpc_data() const noexcept;
```

**Complex math (`mpfrxx` namespace).** `sqrt`, `exp`, `log`, the trigonometric and hyperbolic families and their inverses, `pow`, `conj`, `proj`, and the projections `real`, `imag`, `abs`, `norm`, `arg`. Expression-node arguments are accepted.

**Comparisons.** `mpc_class` supports equality and inequality only; ordering operators are not defined. A value with a NaN component compares unequal to itself.

## 10 Rounding and evaluation contexts

### 10.1 MPFR rounding

`mpfrxx::with_rounding(value, rnd)` returns a lightweight non-owning target-bound handle that applies `rnd` to assignment and compound assignment through that handle. It carries no precision field: the destination `mpfr_class` determines the result precision exactly as a direct `mpfr_*` call would. The handle must not outlive, be stored beyond, or be used after destruction or move of the referenced object. It does not create a dynamic rounding scope and does not affect independently materialized eager math such as `sin(x)` or `const_pi(prec)`.

`mpfrxx::rounding_mode_scope` changes the calling thread's MPFR default rounding mode for the lifetime of the scope, restores the previous value on destruction, and updates the stable-rounding cache. Nested scopes are supported. Eager MPFR math functions and constants observe the current default rounding mode and therefore observe a `rounding_mode_scope`, but they are not affected by target-bound `with_rounding` handles.

### 10.2 MPC contexts

MPC keeps a target-bound precision-and-rounding context because it needs separate real and imaginary precision. `mpfrxx::mpc_evaluation_context` stores real precision, imaginary precision, and an `mpc_rnd_t`; `mpfrxx::with_context(value, context)` returns a non-owning handle. Assignment and compound assignment through the handle evaluate with the cached rounding and target component precisions, avoiding per-operation default lookups in fixed-context loops. The handle may also be built from separate real and imaginary MPFR rounding modes, packed with `MPC_RND(real, imag)`. The handle must not outlive the referenced object; normal builds check that the context precision matches the target.

## 11 Compile-time fast paths

The three fast-path macros are compile-time contracts for hot kernels, enabled deliberately per translation unit or target. They are independent: fixed precision does not imply stable rounding, stable rounding does not imply fused arithmetic, and fused arithmetic implies no fixed-precision contract.

| Macro                                     | Meaning                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>GMPFRXX_MKII_FAST_FIXED_PREC</code> | Applies to <code>mpf_class</code> , <code>mpfr_class</code> , <code>mpc_class</code> . Caller guarantees that all participating floating objects in the hot path share one fixed precision; a mismatch is a contract violation (asserted in debug builds). Enables swap-based move assignment and bounded scratch reuse that skip precision-restoration work. |
| <code>GMPFRXX_MKII_FAST_STABLE_RND</code> | Applies to MPFR (and MPC paths inheriting MPFR rounding). Caller guarantees the default rounding mode is stable during the kernel, except through wrapper setters or <code>rounding_mode_scope</code> . Replaces repeated default-rounding reads with a cached thread-local value.                                                                            |

| Macro                   | Meaning                                                                                                                                                                                                                                                            |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GMPFRXX_MKII_ENABLE_FMA | Applies to MPFR expression and compound-assignment paths. Maps $a += b*c$ to <code>mpfr_fma</code> , $a -= b*c$ to <code>mpfr_fma</code> with an exactly negated multiplicand, $a*b + c*d$ to <code>mpfr_fmms</code> , and $a*b - c*d$ to <code>mpfr_fmms</code> . |

**Numeric-semantics note.** GMPFRXX\_MKII\_ENABLE\_FMA *changes* results: fused evaluation rounds once where a materialized product-then-add rounds twice. This is why it is separate from the fixed-precision macro, which does not change arithmetic results. GMP MPF has no wrapper FMA:  $a += b*c$  on `mpf_class` may use a direct multiply-then-add specialization but still follows MPF’s two-step rounding. Under GMPFRXX\_MKII\_FAST\_FIXED\_PREC, the MPF leaf-product compound path may reuse a bounded header-local thread scratch object for the intermediate product; this is a performance cache only and does not change MPF precision semantics. If a scratch pool is exhausted by a deep expression tree, the implementation falls back to a scoped backend temporary without changing observable results, precision policy, or exception behavior.

## 12 `mpq_class` stream input

Stream extraction for `mpq_class` and raw `mpq_ptr` intentionally matches upstream `gmpxx.h`. It uses `mpq_set_str`-compatible semantics, preserves the parsed numerator and denominator, and does *not* call `mpq_canonicalize`. Input such as 2/4 remains 2/4 at the stream layer, and structurally invalid raw rationals such as 1/0 or 0/0 are accepted by extraction rather than setting `failbit`. The stream layer is a raw serialization compatibility boundary, not an arithmetic-ready one.

Constructors, `set_str`, expression evaluation, and arithmetic-producing paths continue to reject zero denominators and produce canonical rationals. Before passing a stream-loaded value to GMP rational arithmetic, validate and canonicalize it with `mpq_class::canonicalize()` (which first rejects a zero denominator with `std::domain_error`, then calls `mpq_canonicalize`) or `gmpxx::mpq_canonicalize_checked(mpq_ptr)`. If the canonicalization throws, the target retains its original raw stream state and is not arithmetic-ready.

## 13 Move semantics and layout guarantees

Floating wrappers are nothrow move-constructible and add no per-object ownership flag around the backend object. Their size and alignment are part of the performance contract:

```
sizeof(gmpxx::mpf_class) == sizeof(mpf_t);
alignof(gmpxx::mpf_class) == alignof(mpf_t);
sizeof(gmpxx::mpfr_class) == sizeof(mpfr_t);
sizeof(gmpxx::mpc_class) == sizeof(mpc_t);
```

This preserves dense array stride for BLAS-like kernels: an array of `mpfr_class` has the same stride as an array of `mpfr_t`, with no extra padding from a wrapper-side validity flag.

Move construction keeps the moved-from source as a valid backend object by initializing destination storage and swapping; this may allocate but avoids ownership metadata. In normal builds, move assignment to a valid destination preserves the destination precision. Under GMPFRXX\_MKII\_FAST\_FIXED\_PREC, move assignment may assume that source and destination precisions match and take the fast swap path; a mismatch is a caller contract violation.

A moved-from floating object remains valid for destruction, assignment, and swap, but its value and precision are unspecified until reassigned. Portable code must not read a moved-from

wrapper for numeric meaning or precision before assigning a new value. `mpfc_class` follows MPF move behavior through its two components.

## 14 Platform notes

**Windows MPC division.** On `_WIN32` builds, `mpc_class` division may use a scaled finite-value workaround instead of calling `mpc_div` directly. It is a portability guard for MinGW/MPC builds where a direct division can overflow near MPFR’s 32-bit-style exponent limits even when the exact quotient is finite and small. The workaround is narrow: it applies only when all components are finite, the divisor is not  $0 + 0i$ , and at least one component’s exponent is within a small margin of the current MPFR `emin/emax`. It performs componentwise max-scaling in MPFR temporaries and then rounds with the requested MPC rounding mode. It is a compatibility path, not a separate complex-division semantics; last-bit differences from a direct `mpc_div` are possible. Non-Windows builds call `mpc_div` directly.

## 15 Benchmark results

The repository contains reproducible benchmark reports under `benchmarks/gmp/*/README.md` and `benchmarks/mpfr/*/README.md`. Those reports are the authoritative source for raw logs, CSV files, plots, hotpath disassembly, exact command lines, and per-variant interpretation. This section records representative headline results from the committed benchmark data used for the 1.0.0 release-candidate documentation.

The 00/01/02 vector and matrix-vector rows use the repeat-10 runs recorded by `benchmarks/run_all.sh`. The Rgemm rows use the clean 512-bit pow2/64n sweep recorded by `benchmarks/run_rgemm_all.sh`. MFLOPS values are elapsed-time metrics, not hardware-counter measurements. In the table below, entries of the form “Max / Avg” report maximum and average MFLOPS over repeats. Rgemm rows show the representative large-size maximum at  $N = 1472$ , where most sizes were measured with `repeat=1`.

| Backend | Benchmark | Case              | Best serial representative            | Best OpenMP representative         |
|---------|-----------|-------------------|---------------------------------------|------------------------------------|
| GMP     | Rdot      | 512-bit           | mkII_06_FIXED, 33.796 / 32.961        | omp_mkII_03, 578.634 / 573.121     |
| GMP     | Rdot      | 1024-bit          | mkII_06_FIXED, 12.151 / 11.897        | omp_mkII_03, 323.366 / 315.184     |
| GMP     | Raxpy     | 512-bit           | C_native_03, 34.040 / 33.727          | omp_orig_02, 398.110 / 393.715     |
| GMP     | Raxpy     | 1024-bit          | C_native_01, 12.042 / 11.970          | omp_orig_02, 255.925 / 252.424     |
| GMP     | Rgemv     | 512-bit           | mkII_03, 31.991 / 31.449              | C_omp_07, 552.697 / 540.759        |
| GMP     | Rgemv     | 1024-bit          | mkII_03_FIXED, 11.586 / 11.268        | C_omp_07, 264.362 / 262.194        |
| GMP     | Rgemm     | 512-bit,<br>=1472 | N C_native_02, 31.346                 | C_omp_06, 871.995                  |
| MPFR    | Rdot      | 512-bit           | C_native_06, 23.936 / 23.688          | C_omp_01_FMA, 600.518 / 560.417    |
| MPFR    | Rdot      | 1024-bit          | mkII_05_RPF, 10.425 / 10.183          | C_omp_01_FMA, 293.591 / 286.659    |
| MPFR    | Raxpy     | 512-bit           | mkII_01_RPF, 23.209 / 22.963          | C_omp_01_FMA, 422.804 / 415.811    |
| MPFR    | Raxpy     | 1024-bit          | C_native_01_FMA, 9.393 / 9.351        | omp_mkII_01_RPF, 256.399 / 252.310 |
| MPFR    | Rgemv     | 512-bit           | mkII_02_ROUNDING_FMA, 23.959 / 23.408 | omp_mkII_07_RP, 455.524 / 443.905  |
| MPFR    | Rgemv     | 1024-bit          | mkII_04_RPF, 10.434 / 10.152          | C_omp_07_FMA, 239.287 / 233.270    |
| MPFR    | Rgemm     | 512-bit,<br>=1472 | N C_native_02, 24.069                 | C_omp_02, 686.758                  |

Table entries are shortened for page fit. Prefix `omp_` denotes an OpenMP target, and `c_` denotes a raw C native target. The shortened suffix `FIXED` denotes the fixed-precision fastpath build (`FIXED_PRECISION_FASTPATH`). The shortened MPFR suffix `RPF` denotes the `ROUNDING_PRECISION_FMA` class. Full executable names, sorted tables, raw-data paths, plots, and disassembly snippets are intentionally kept in the benchmark README files rather than duplicated here.

**Interpretation.** The committed data show three stable boundaries. First, reusable temporaries and fixed-precision scratch paths control most GMP wrapper overhead in expression benchmarks. Second, MPFR wrappers need loop-external rounding plus an FMA-capable source shape before their hot loop reaches the raw C FMA call class. Third, Rgemm is not primarily a wrapper problem in the current sweep: at N=1472, the best GMP and MPFR wrappers are close to their raw C backend class, while GMP remains about 1.3 times faster than MPFR in the best serial C native kernels.

The benchmark reports should be consulted before drawing narrower conclusions. They include static sorted tables, line plots, and representative disassembly showing whether `mpf_init2` / `mpfr_init2`, rounding-mode lookup, temporary construction, `mpfr_fma`, and OpenMP final reductions appear in or outside the timed hot loop.

## 16 Version information

```
const char* gmpxx::version();           // also mpfrxx::version()
const char* gmpxx::git_commit_hash();    // also mpfrxx::git_commit_hash()
std::ostream& gmpxx::print_version(std::ostream&);
std::ostream& gmpxx::print_git_commit_hash(std::ostream&);
```

## 17 License

**gmpfrxx\_mkII** is distributed under the BSD 2-Clause license, Copyright © 2026 Nakata, Maho. The full text appears in the header of each source file.